

NLP Question Bank

100 Questions

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Evaluation Metrics (Precision, Recall, F1)

1. In relation extraction, given gold and predicted tuples, compute Precision, Recall, and F1.
2. If Precision = 0.9 and F1 = 0.9, what is Recall?
3. Explain why F1-score can be more informative than accuracy in imbalanced datasets.
4. Given TP = 90, FP = 30, FN = 10, TN = 200, compute Precision, Recall, F1.
5. Model A: P = 0.8, R = 0.4; Model B: P = 0.6, R = 0.6. Which has higher F1?
6. Discuss advantages and limitations of distant supervision.

Minimum Edit Distance (Levenshtein)

7. Find the minimum number of edits to transform “COMPUTE” into “COMMUTE”. Show alignment.
8. For the strings “INTENT” and “CONTENT”, trace back one optimal alignment with all costs = 1.
9. Compute the edit distance between “INTENTION” and “EXECUTION” assuming unit cost operations. Show the DP matrix and optimal edits.
10. Compute the minimum edit distance between “COMPUTE” and “COMMUTE” with insertion, deletion, substitution each cost 1. Show the DP table.
11. For the strings “HONEST” and “HONOR”, find minimum edit distance with substitution cost 2, indels cost 1.
12. Given strings “NATURAL” and “NEUTRAL”, find edit distance with insertion/deletion cost 1 and substitution cost 2. Explain the difference.

HMM for POS Tagging

13. Compute probability of tag sequence “NN VB” for “dogs bark” using given emission and transition probabilities.
14. Why is the Viterbi algorithm used instead of enumerating all tag sequences?
15. Derive the forward probability for a 2-step HMM sequence.
16. In the same HMM, if $P(\text{JJ} \rightarrow \text{JJ}) = 0.2$ and $P(\text{JJ emits “light”}) = 0.1$, what is the probability of “JJ JJ” generating “light light”?
17. Explain HMM components for POS tagging. Given: $P(\text{START} \rightarrow \text{JJ}) = 0.5$, $P(\text{JJ} \rightarrow \text{NN}) = 0.3$, $P(\text{JJ emits “bright”}) = 0.4$, $P(\text{NN emits “light”}) = 0.7$, compute probability of tag sequence “JJ NN” generating “bright light”.
18. Given transition probabilities, compute likelihood of a tag sequence for a 2-word sentence.

Chu-Liu-Edmonds MST Algorithm

19. Describe cycle contraction in MST algorithms with an example.
20. Explain dependency parsing using graph-based methods.
21. Explain Chu-Liu-Edmonds algorithm for MST dependency parsing. Given root R and nodes X, Y, Z with edge weights: $R \rightarrow X = 10, R \rightarrow Y = 9, R \rightarrow Z = 8, X \rightarrow Y = 7, X \rightarrow Z = 6, Y \rightarrow X = 5, Y \rightarrow Z = 4, Z \rightarrow X = 3, Z \rightarrow Y = 2$, find the maximum spanning tree. Which cycle is contracted first?
22. Why is the algorithm guaranteed optimal for directed graphs?

Bigram MLE & Perplexity

23. From the same corpus, compute perplexity of the bigram model on the test sentence “a bird sings”.
24. Define maximum likelihood estimation for bigram probabilities. From the corpus: “a bird is on the tree”, “the bird sings”, “a tree is tall”, compute $P(\text{bird} \mid \text{a})$ and $P(\text{tree} \mid \text{the})$.
25. Compute $P(\text{slow} \mid \text{runs})$ from the same corpus.
26. Explain why perplexity is used to evaluate language models. A model assigns probabilities 0.4, 0.3, 0.2, 0.1 to four test words. Compute perplexity.
27. Derive unigram and bigram probabilities from a toy corpus and compare.
28. Given a corpus: S1: “a dog runs fast”, S2: “a cat runs slow”. Compute $P(\text{runs} \mid \text{dog})$ using MLE.
29. Calculate perplexity of a bigram model for the sentence “a dog runs”.
30. Explain how data sparsity affects N-gram models with examples.

Laplace (Add-k) Smoothing

31. Explain Add-k smoothing. Compute probability for an unseen bigram when $k = 0.5$, $V = 8000$, $\text{count}(\text{deep})=300$, $\text{count}(\text{deep, learning})=0$.
32. Why does smoothing improve generalization in probabilistic models?
33. For a bigram model with $V = 5000$, $\text{count}(\text{the})=200$, $\text{count}(\text{the, cat})=0$, compute the difference between smoothed and unsmoothed $P(\text{cat} \mid \text{the})$.

34. Why is add-1 smoothing needed? For vocabulary $V = 8000$, “apple” appears 250 times, “apple pie” never appears. Compute $P_{\text{Add-1}}(\text{pie} \mid \text{apple})$.
35. Given $\text{count}(\text{go}, \text{home})=12$, $\text{count}(\text{go})=40$, $V = 6000$, compare MLE vs add-1 smoothed probability.
36. What is the main disadvantage of add-1 smoothing when V is large? Illustrate.

Latent Dirichlet Allocation (LDA)

37. Why can increasing the number of topics K lead to overfitting? Discuss.
38. Explain Gibbs sampling for topic assignment.
39. What happens when Dirichlet priors become very small?
40. For document d : $n_1^{(d)} = 20$, $n_2^{(d)} = 10$, $n_3^{(d)} = 5$, total assignments=50, $K = 5$, $\alpha = 0.2$. Compute $\theta_1^{(d)}$ and $\theta_2^{(d)}$.
41. Explain the role of hyperparameters α and β in LDA. Suppose $\alpha = 0.2$, $\beta = 0.05$. If both decreased to 0.01, how does sparsity change?
42. Explain topic modeling using probabilistic methods.
43. Describe the role of hyperparameters in topic models.
44. Using Gibbs sampling: For topic k , word “algorithm” count $n = 8$, total words in topic =150, $V = 8000$, $\eta = 0.1$. Compute $\beta_{\text{algo}}^{(k)}$.

Cosine Similarity & TF Vectors

45. Compute cosine similarity between vectors $A = (3, 1, 2)$ and $B = (1, 4, 0)$.
46. Which pair of words is most similar? Justify.
47. Explain Euclidean vs cosine similarity differences.
48. Given document vectors (TF only): Word “Network”: Doc1=5, Doc2=2, Doc3=1; “Security”: 3,4,2; “Data”: 1,3,5. Compute cosine similarity (a) between words “Data” and “Security”, (b) between Doc1 and Doc2.
49. Given vectors, determine which pair is closest.
50. Why is cosine similarity preferred over Euclidean distance for document similarity?

Zipf’s Law & Heaps’ Law

51. A corpus has 15,000 unique words after 2 million tokens. Using Heaps’ law with $k = 25$, estimate the exponent β .
52. Explain why Heaps’ law predicts sub-linear vocabulary growth. For $k = 50$, $\beta = 0.4$, find V when $N = 10^6$ and when $N = 10^7$.
53. If a corpus follows Zipf’s law exactly, compute the ratio $f(10)/f(20)$.
54. Using vocabulary growth laws, estimate the lexicon size change when corpus increases from 2 million to 8 million tokens, given $k = 30$, $\beta = 0.6$.
55. Describe the rank-frequency distribution in text. If the most frequent token occurs 10,000 times, estimate the frequency of the 25th ranked token.
56. Derive the mathematical relationship between word frequency and rank. If the most frequent word appears 8,000 times, what is the frequency of the 40th ranked word?
57. State Heaps’ Law. A corpus follows Heaps’ law with constants $k = 30$ and $\beta = 0.45$. Predict vocabulary size when corpus size grows from 500,000 to 2,000,000 tokens.
58. Explain Zipf’s Law with its mathematical form. In a corpus, the most frequent word occurs 12,000 times. Predict the frequency of the 25th most frequent word.

Chomsky Normal Form (CNF)

59. Explain ambiguity in CFG with an example.
60. Why is CNF important for parsing algorithms like CYK?
61. Define Chomsky Normal Form. Convert: $S \rightarrow ABC$, $A \rightarrow a \mid Aa$, $B \rightarrow b$, $C \rightarrow c \mid Cc$ to CNF.
62. Define normal forms in grammar representation and explain their significance.

Probabilistic CFG (PCFG)

63. Compute total probability of all parses for a sentence.
64. Explain why PCFGs can assign higher probability to shorter trees. How does this affect parsing?
65. Given a PCFG, calculate the probability of a parse tree.
66. Given PCFG rules: $S \rightarrow NP VP$ (0.7), $S \rightarrow VP$ (0.3); $NP \rightarrow N$ (1.0); $VP \rightarrow V$ (1.0); $N \rightarrow \text{dog}$ (0.6), $N \rightarrow \text{barks}$ (0.4); $V \rightarrow \text{dog}$ (0.3), $V \rightarrow \text{barks}$ (0.7). For “dog barks”: (i) number of parse trees, (ii) probability of each, (iii) total probability.

Skip-gram & Word Embeddings

67. For a sentence of length 8 and window size 3, compute the number of training pairs.
68. Explain CBOW architecture and compare with skip-gram.
69. Explain negative sampling technique.
70. Describe skip-gram architecture. With $V = 100,000$, $d = 200$, how many parameters? For sentence “machine learning is fun” with window size $c = 2$, list all (center, context) pairs.
71. A skip-gram model with $V = 150,000$ and $d = 250$ has how many parameters? If V is halved and d doubled, does the parameter count increase or decrease?
72. Explain why skip-gram uses two separate embedding matrices.
73. What is the effect of increasing embedding dimension d on model capacity and training time?
74. For vocabulary size 40,000 and embedding dimension 200, calculate the number of parameters.

Maximum Entropy (MaxEnt) Principle

75. How does MaxEnt differ from Naive Bayes? What constraints can it incorporate?
76. Explain the Maximum Entropy principle in classification problems.
77. Compare MaxEnt models with Naive Bayes classifiers.
78. State the maximum entropy principle for classification. A MaxEnt POS tagger has $V = 5000$, $\langle \text{UNK} \rangle$ probability = 0.06, 100 closed-class words share total probability 0.40. All other words equally likely. Probability of unseen open-class word “transform”?

KL Divergence & Jaccard Similarity

79. If Jaccard similarity = 0.2 and $|A \cup B| = 50$, find $|A \cap B|$.
80. Compute Jaccard similarity between binary vectors: $A = [1, 0, 1, 0, 1, 1, 0]$, $B = [0, 1, 1, 0, 1, 0, 1]$.
81. Given $p = [0.5, 0.3, 0.1, 0.1]$, $q = [0.2, 0.5, 0.2, 0.1]$, compute $D(p||q)$ and $D(q||p)$ (log base 2). Which is larger? Explain asymmetry.
82. Compute KL divergence between two distributions using base 2 logs.
83. For distributions $p = [0.7, 0.3]$, $q = [0.5, 0.5]$, compute KL divergence in bits. Is it symmetric?
84. Explain why KL divergence is asymmetric with an example.

Distant Supervision & Relation Extraction

85. Compare supervised vs weakly supervised extraction techniques.
86. How can you reduce false positives in distantly supervised relation extraction?
87. Define distant supervision in information extraction.
88. Explain distant supervision. Compare with bootstrapping (e.g., DIPRE, Snowball). Give one advantage and one limitation of distant supervision.
89. Given extracted tuples: TP=3, FP=1, FN=2. Compute Precision, Recall, F1.
90. Calculate F1-score and explain when it is preferred over accuracy.

Additional Mixed Problems

91. In LDA, if α is very large, what distribution of topics per document do you expect?
92. For the Chu-Liu-Edmonds algorithm, explain cycle contraction with a small example.
93. Given vectors $A = [2, 1, 3]$, $B = [4, 2, 6]$, compute cosine similarity. Interpret.
94. For “HONEST” and “HONOR”, find minimum edit distance with substitution cost 2, indels cost 1.
95. In a PCFG, if a sentence has two parse trees with probabilities 0.3 and 0.5, what is the total sentence probability?
96. Compute perplexity of a bigram model on a 3-word test sentence with probabilities $[0.5, 0.2, 0.3]$.
97. Using add-1 smoothing with $V = 5000$, $\text{count}(\text{the})=200$, $\text{count}(\text{the}, \text{cat})=0$, compute the difference between smoothed and unsmoothed $P(\text{cat}|\text{the})$.
98. A skip-gram model with $V = 150,000$, $d = 250$: compute parameters. Compare with halving V and doubling d .
99. For $p = [0.7, 0.3]$, $q = [0.5, 0.5]$, compute KL divergence. Is it symmetric?
100. In distant supervision, why does aligning a knowledge base with a large text corpus generate noisy labels?